

Regression Analysis Using Linear Regression

Codveda Technology – Data Analytics Internship

Submitted By

Name: Pawan Kushwah

College: G H Raisoni International Skill Tech University, Pune

Branch: B.Tech Artificial Intelligence & Data Science

Objective

The objective of this project is to build a Linear Regression model to predict house prices using the Boston Housing dataset. The model is evaluated using Mean Squared Error (MSE) and R^2 Score.

Dataset

Dataset Name: Boston Housing Dataset

The dataset contains information about various factors affecting house prices, such as crime rate, average number of rooms, pollution level, tax rate, and other housing attributes.

Tools Used

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
-

Methodology

Step 1 – Data Loading

The dataset was loaded into a Pandas DataFrame and column names were assigned.

Step 2 – Data Exploration

The dataset was explored using `head()`, `info()`, and `describe()`.

Step 3 – Feature Selection

The **PRICE** column was selected as the target variable, while the remaining columns were used as input features.

Step 4 – Train-Test Split

The dataset was divided into training (80%) and testing (20%) sets.

Step 5 – Model Training

A Linear Regression model was trained using the training dataset.

Step 6 – Prediction

House prices were predicted using the test dataset.

Step 7 – Model Evaluation

The model performance was evaluated using Mean Squared Error (MSE) and R² Score.

Results

Mean Squared Error (MSE): 24.29

R² Score: 0.669

The model was able to explain approximately **67%** of the variation in house prices.

Conclusion

The Linear Regression model successfully predicted house prices with satisfactory performance. The evaluation metrics indicate that the model provides reliable predictions and demonstrates the practical application of regression techniques in predictive analytics.